

AI for Transportation Systems Planning and Analysis

An Introduction in Concept and Practice

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Alexander Yoshizumi, Kshitiz Khanal, Si Shi

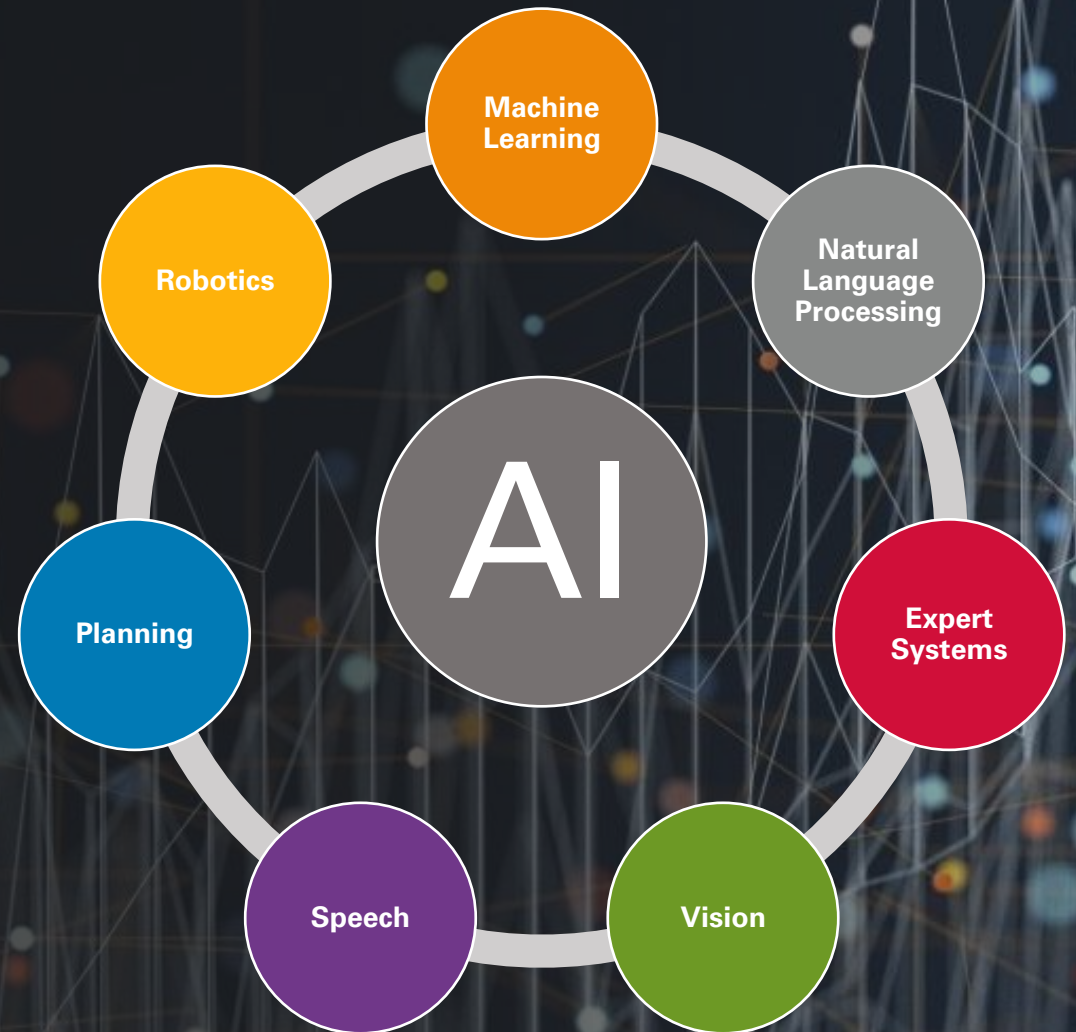
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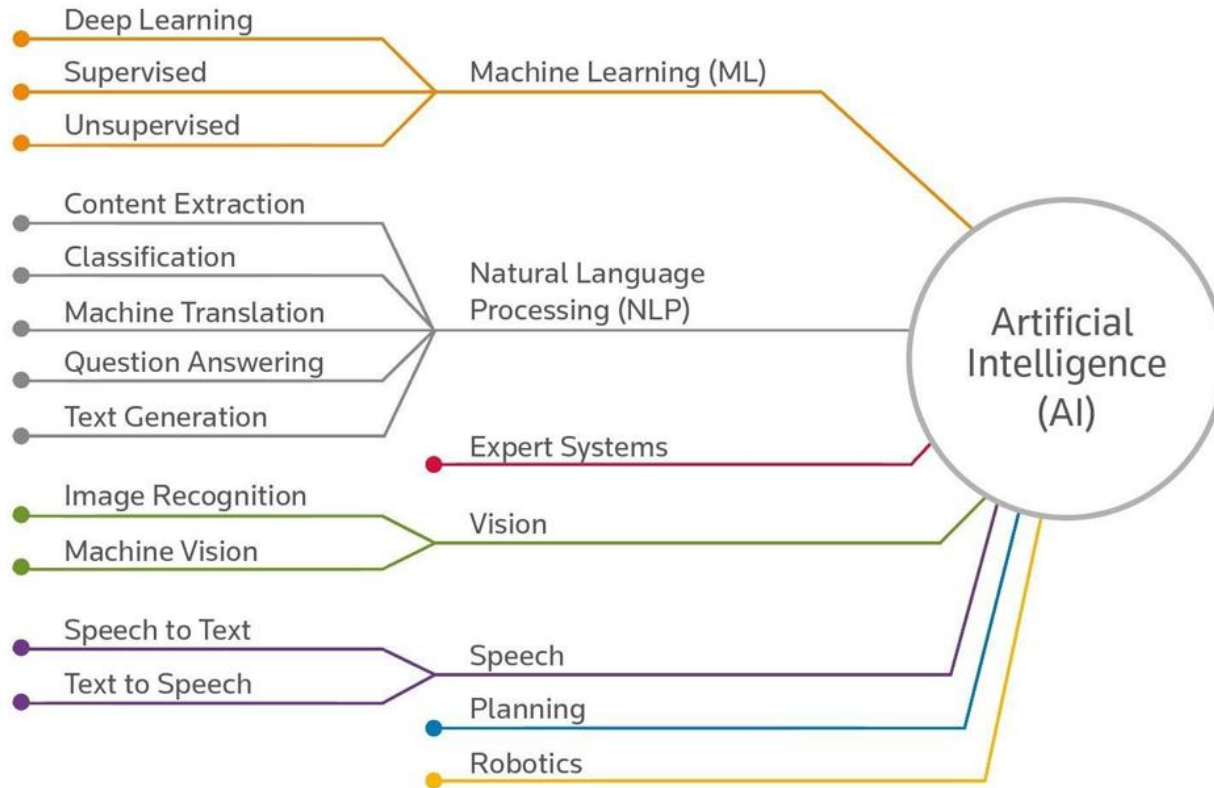
**SYSTEMS
PLANNING &
ANALYSIS**

What Do We Mean When We Say AI?

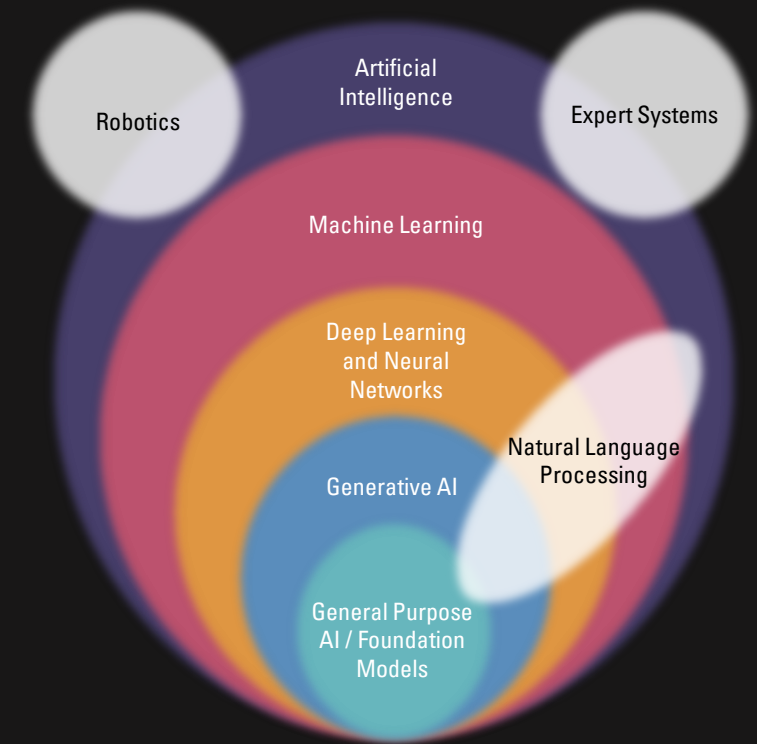
1. All AI methods, not just LLMs.
2. Special emphasis on machine learning, natural language processing, and vision.
3. Opportunities for LLMs and agentic pipelines to support and enhance workflows.



We Can Taxonomize AI in Different Ways



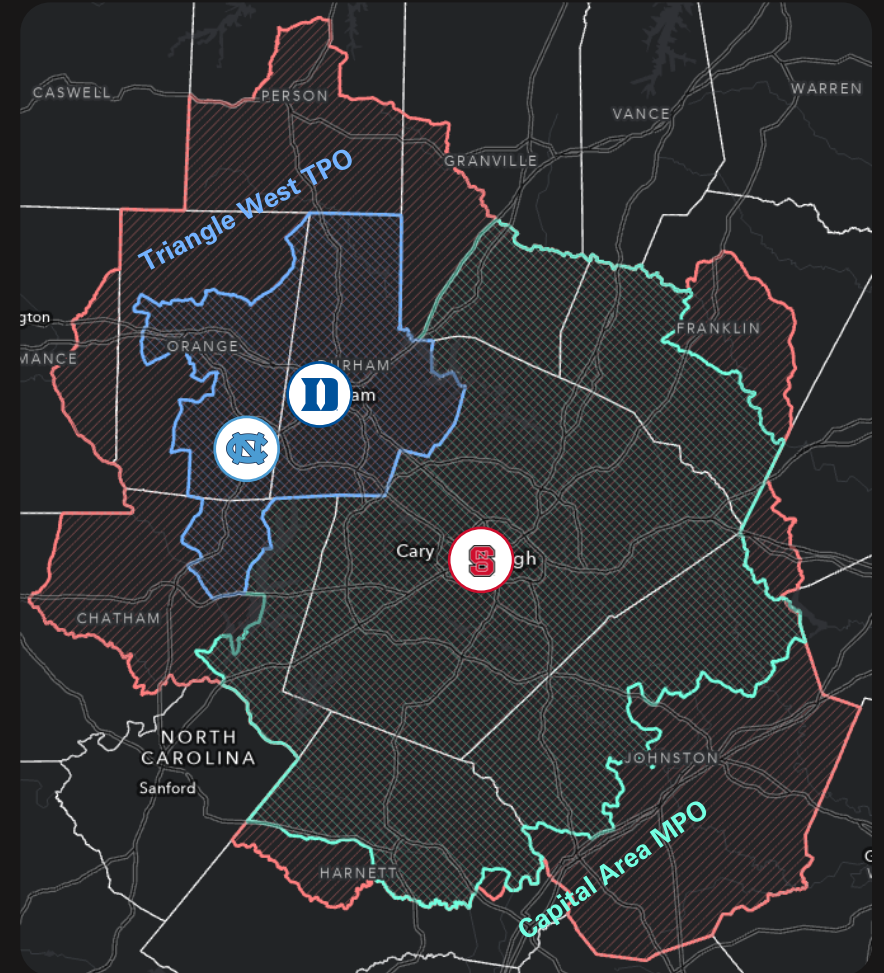
Based on Congressional Research Service Infographic
(<https://www.congress.gov/crs-product/IG10077>)



The Triangle Region of North Carolina

- Raleigh-Durham-Chapel Hill metropolitan area:
 - Intensely multi-nucleated
 - Cary, Apex, Wake Forest, Fuquay-Varina
 - Multiple major research institutions
 - Research Triangle Park
- 2 MPOs – 1 model
- Modeled area
 - >2 million people
 - ~813,000 households

Triangle Regional Model G2 v2

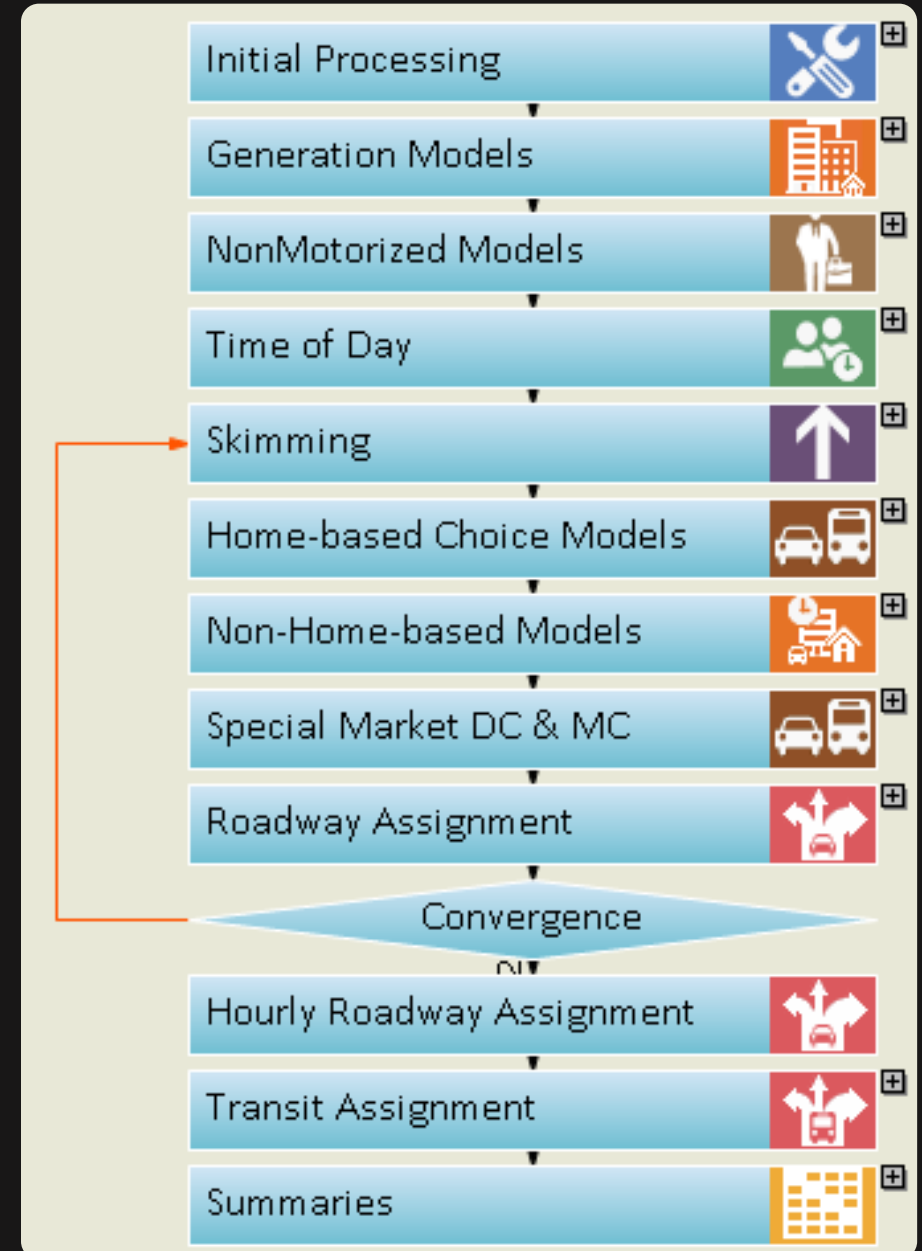


Overview of TRMG2 v2

- Hybrid, trip-based model
- G2 first released in 2022
- ~813,000 HHs
- Covers nearly ~3,933 square miles and 3,147 TAZs.
- Lives on GitHub. Check it out using the QR code below

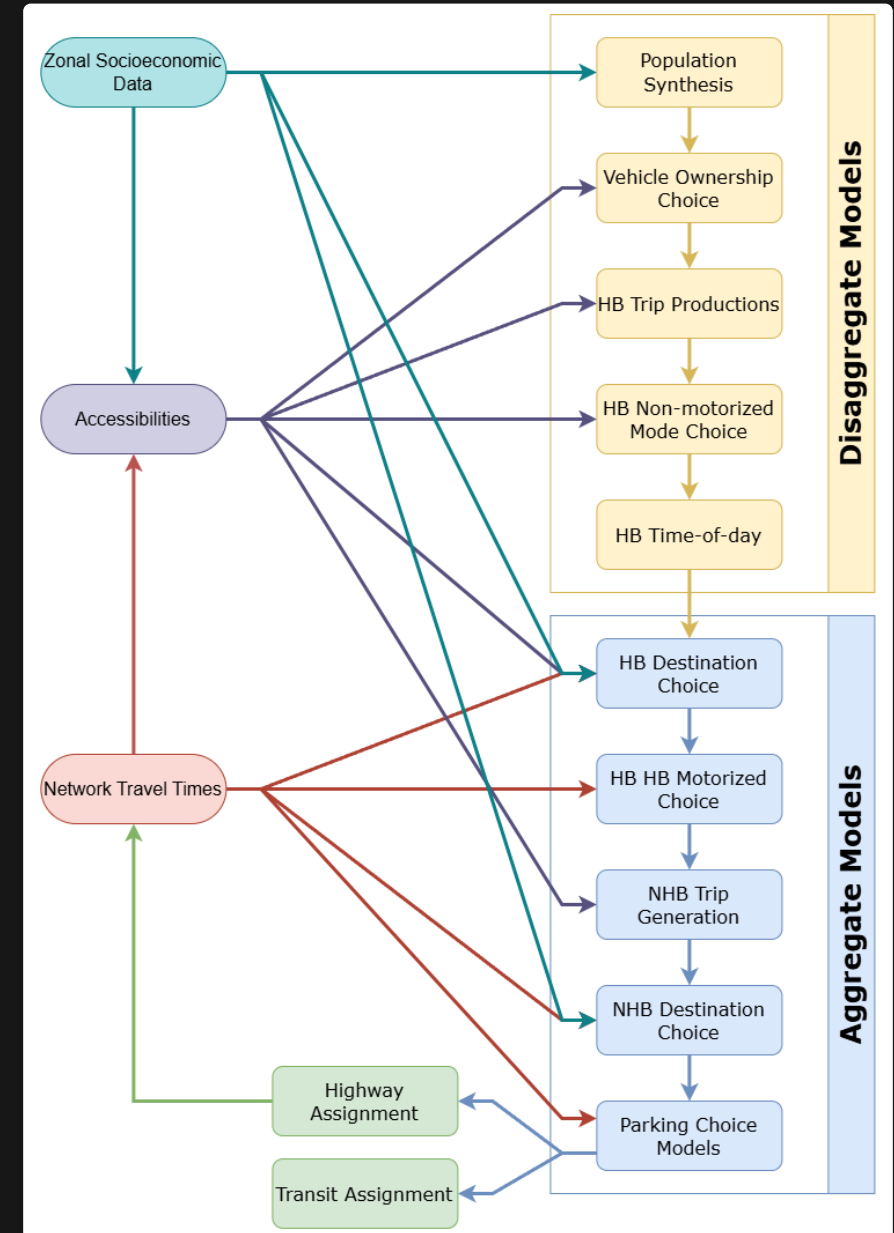


<https://go.ncsu.edu/trm>

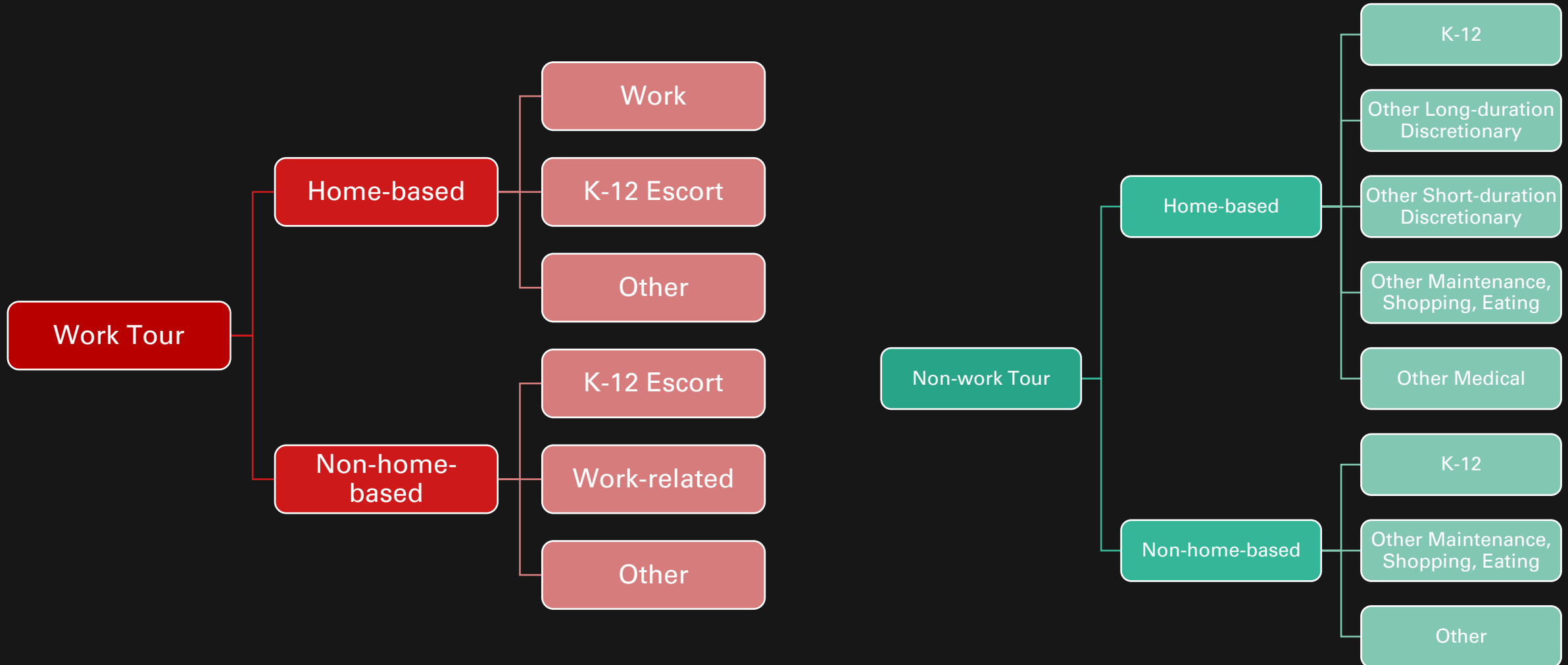


Hybrid Model Architecture

- Disaggregate trip generation, aggregate trip distribution
- Trip-based, segmented by tour and trip type
- Machine learning components
 - Decision trees in Trip Generation
 - Boosting in NHB Destination Choice



Trip Productions by Tour Type and Trip Type



Decision Trees for Trip Generation

- Cross-compared different statistical and machine learning methods.
- Chose simple, ANOVA-based Decision Trees.
 - **Which variables best divide the population into distinct groups?**
 - Interpretable, stronger predictive power, captures non-linear effects

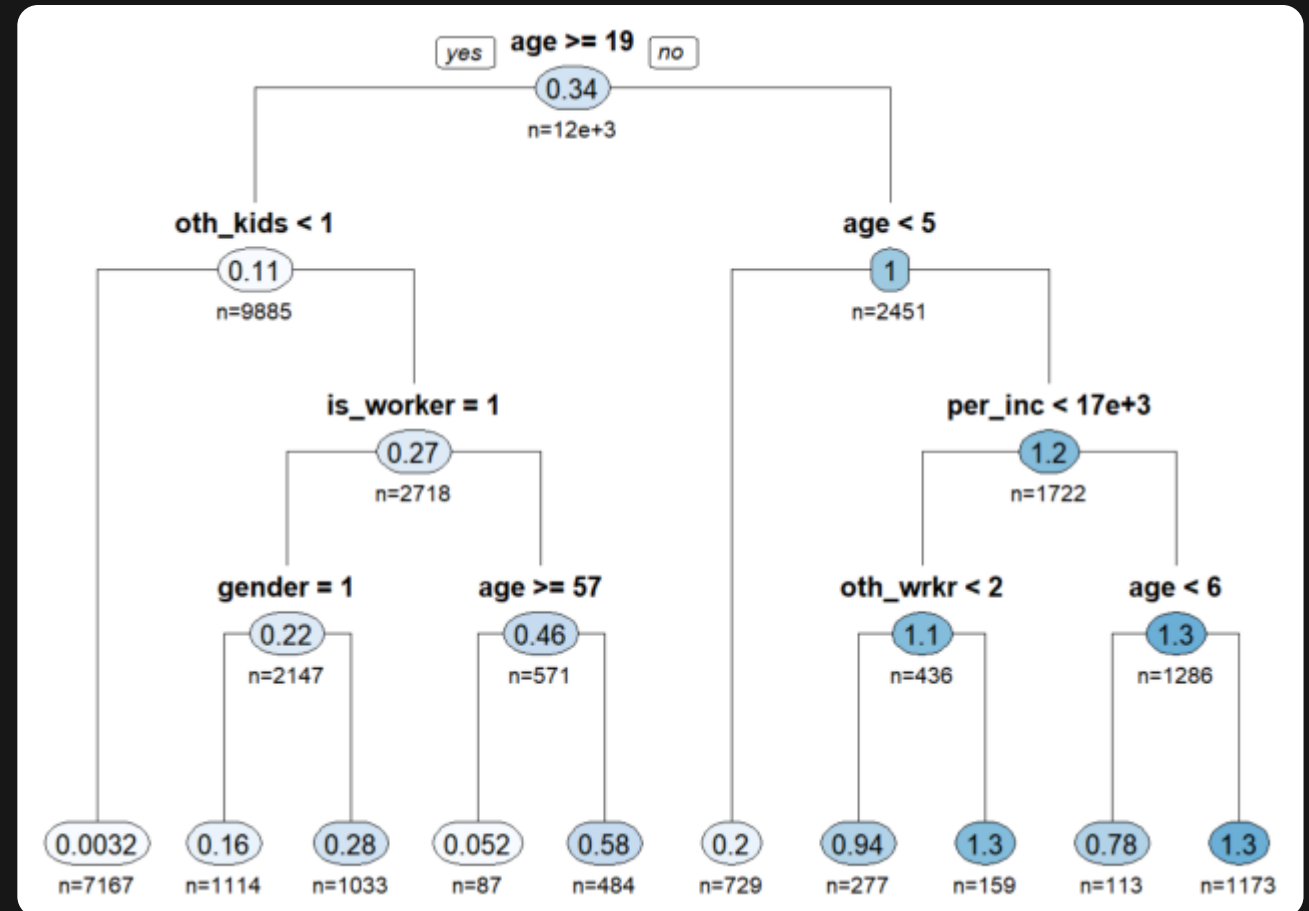
Example: School Trips

Model Type	Pseudo R ²
Ordered Logit	0.03
GLM (Regression)	0.22
Cross-Classification	0.33
XGBoost	0.60
Simple Decision Tree	0.53

Decision Trees for Trip Generation

Example: School Trips

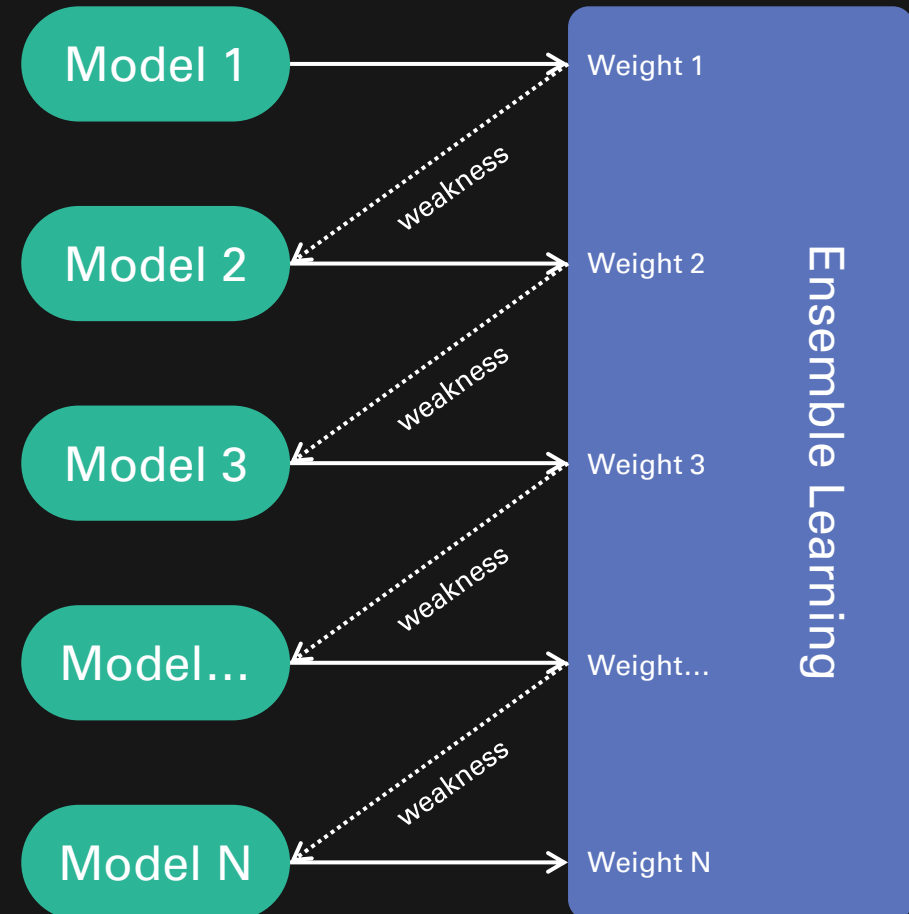
Model Type	Pseudo R ²
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Average Trip Rate and Number of Observations at Each Split

Boosting for Non-Home-Based Trips

- Boosting is an ensemble machine learning technique where multiple models are run in sequence, and the errors of each preceding model are used to estimate each following model.

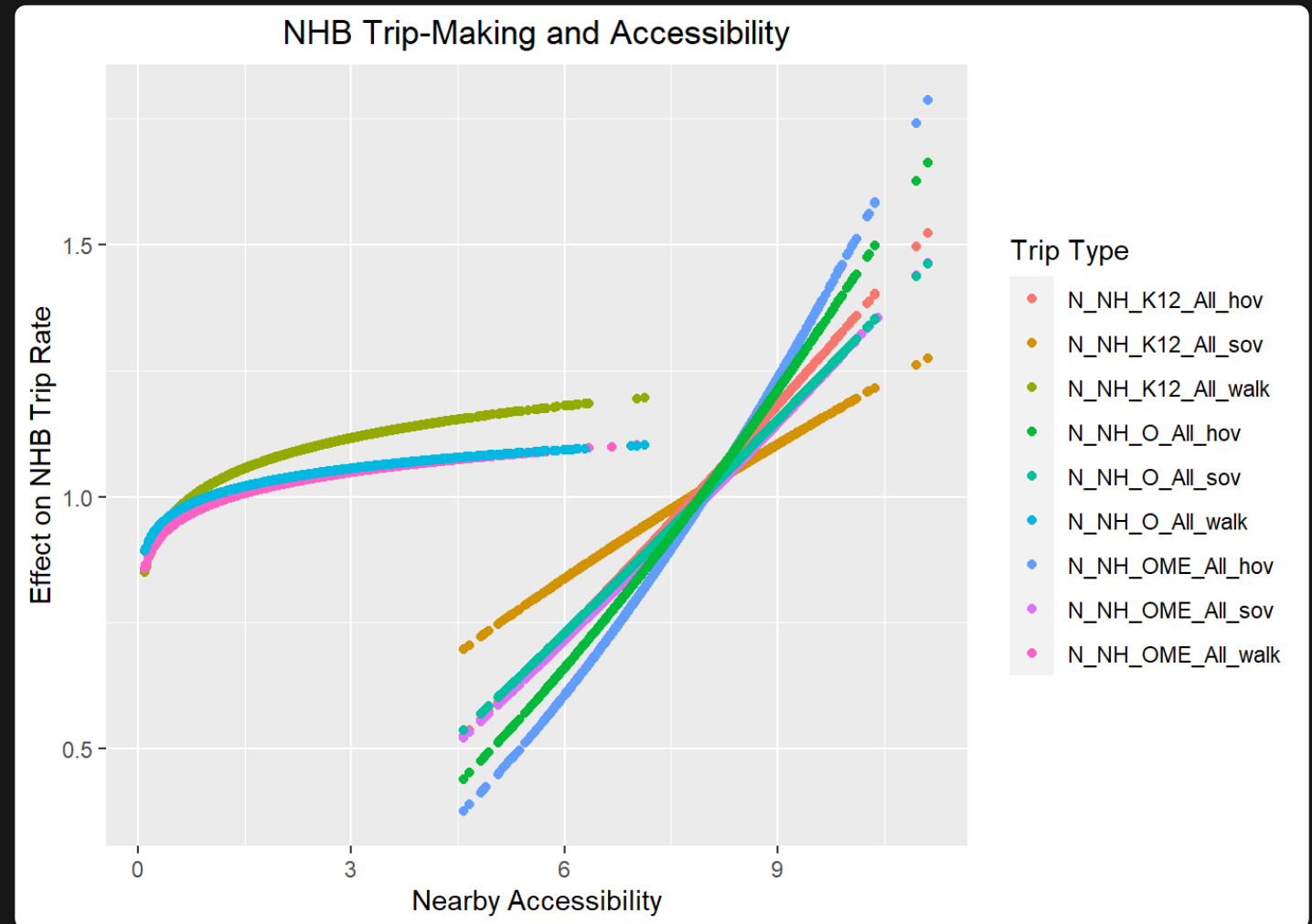


Boosting for Non-Home-Based Trips

$$Y_{p,m} = \alpha A^\gamma \sum_{t,m} \beta_{t,m} X_{t,m}$$

Where:

- $Y_{p,m}$ is the number of NHB trips of a given NHB trip type (p) and mode (m).
- β is the prior fit coefficient by HB trip type (t) and mode (m).
- $X_{t,m}$ is the number of HB destinations by HB trip type (t) and mode (m).
- A is a measure of accessibility to nearby destinations.
- α and γ are parameters derived from log-log regression on residuals of the log transform of the preceding model

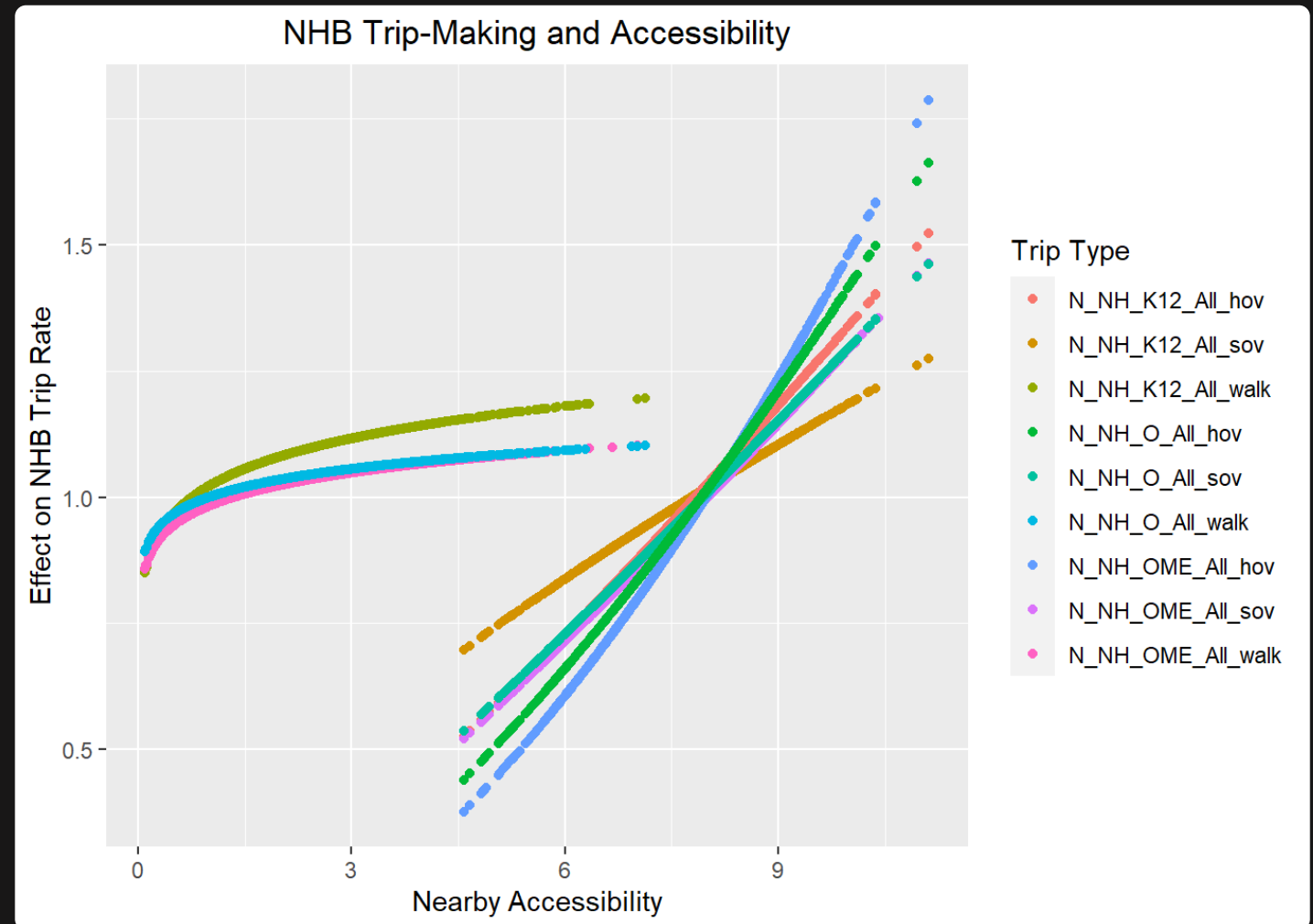


Boosting for Non-Home-Based Trips

$$Y_{p,m} = \alpha A^\gamma \sum_{t,m} \beta_{t,m} X_{t,m}$$

Key Takeaways:

- The accessibility term (αA^γ) derived through boosting scales the effect on the non-home-based trip rate ($Y_{p,m}$).
- The NHB trip rate is decreased (~50%) in rural areas.
- The NHB trip rate substantively increases in more accessible areas.
- Walk accessibility has a more modest effect as walk-based accessibility increases.
- NHB by walk asymptotically approaches 0 in non-walkable areas.



When to Use Machine Learning Methods



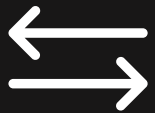
When prediction accuracy is paramount.



When complex, non-linear relationships are present.



When observations counts ($n = \dots$) are large or data are high-dimensional (many features relative to observations).



When many interaction effects are present between variables.

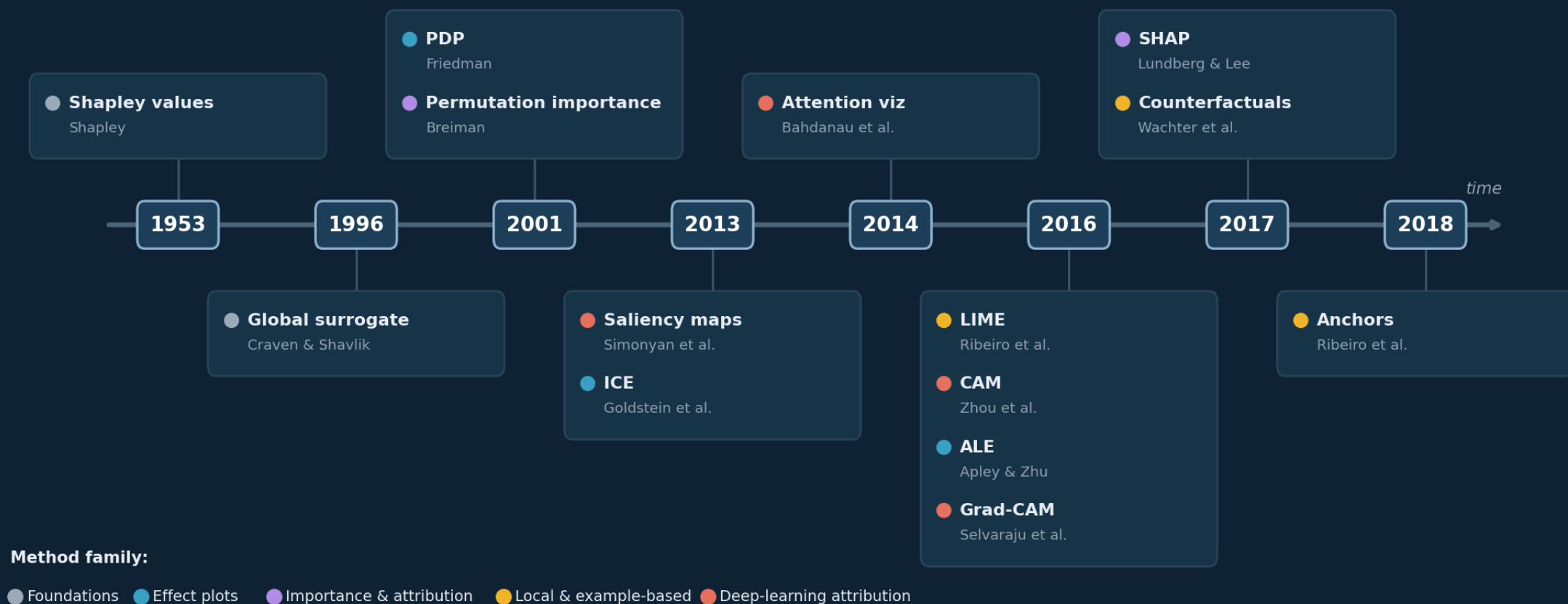
Things to Remember About Machine Learning

- Machine learning is a family of methods, not a single technique.
- Every method encodes assumptions (so you need to match the model architecture to the problem structure).
- Machine learning trades interpretability for flexibility on a spectrum (not all methods are equally opaque).
- Machine learning learns patterns, not truth. An algorithm will faithfully convey the signal in your data, but whether that signal is representative requires analytical rigor to determine.
- Rigorous validation and robustness testing are critical.
- Data quality and quantity remain the binding constraint (not model architecture).

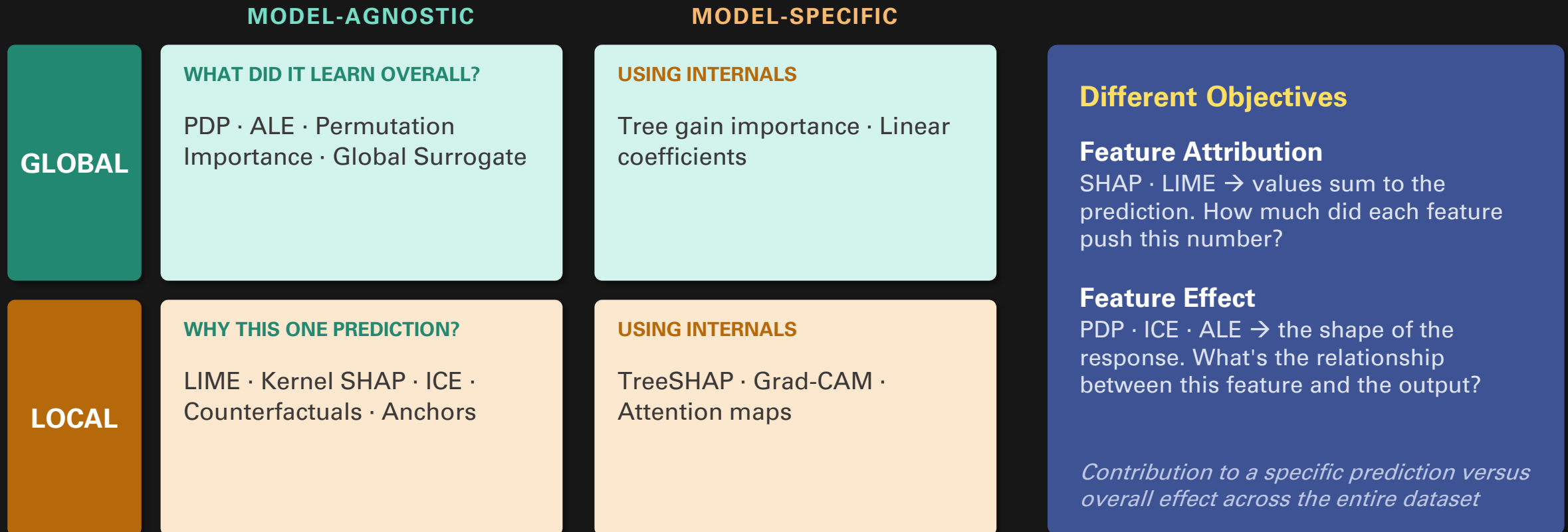
XAI Data Visualizations Continue to Expand

A Timeline of Explainable-AI Visualizations

Year each method was first introduced (first public appearance; peer-reviewed venue in the references).



Quadrant Map for XAI Data Visualization Uses



Example Analysis | Trips Not Taken

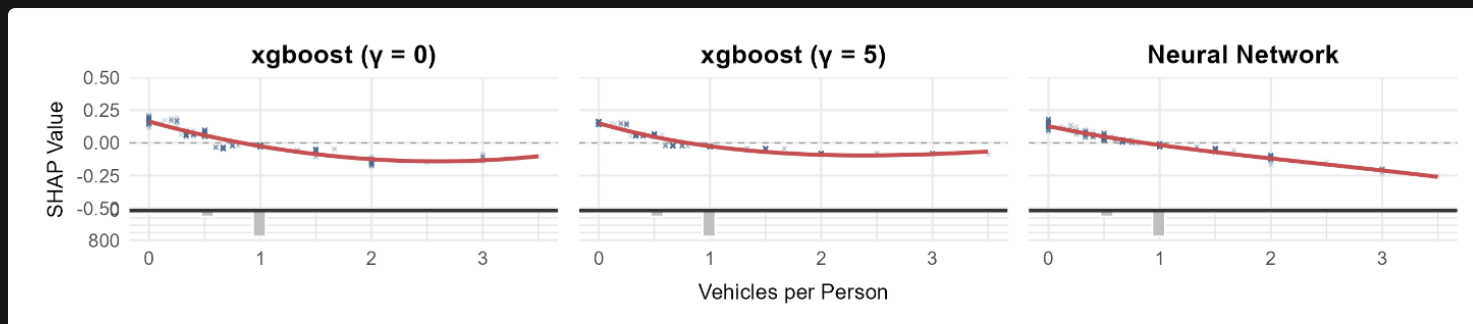
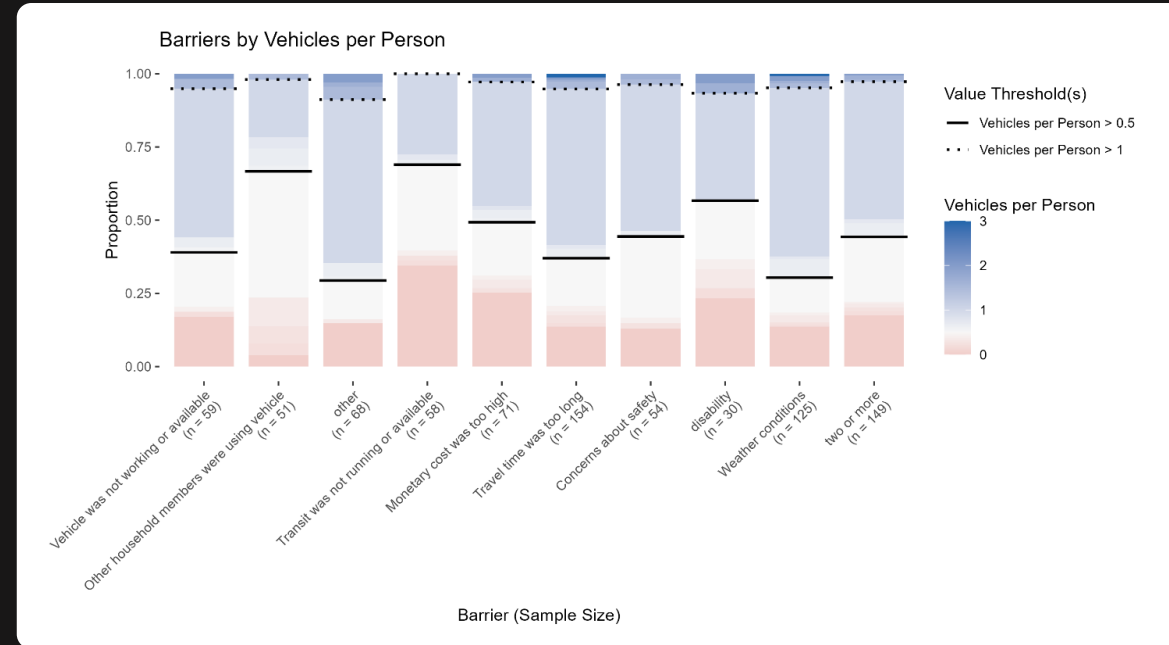
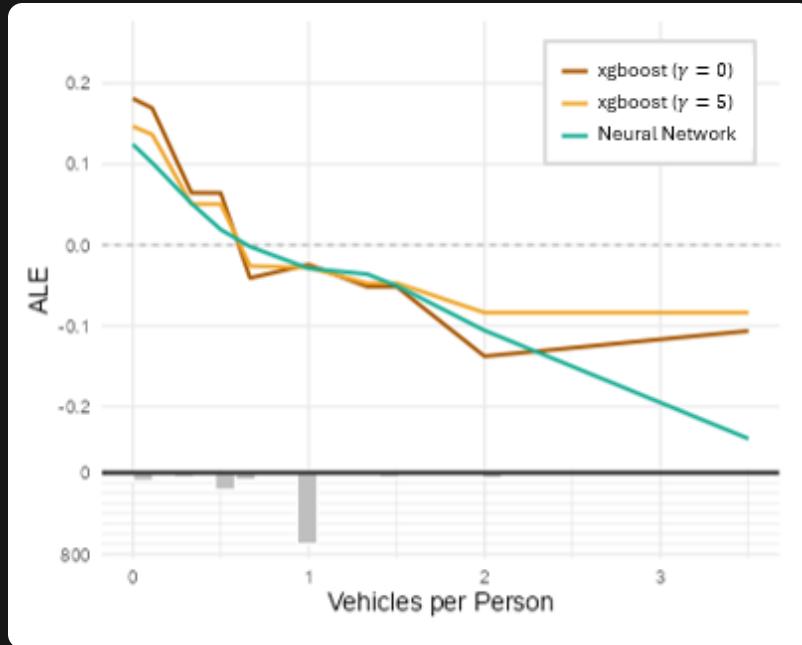
Variable	Directionality	Binomial Logit	Generalized Additive Model	
	XGB1 XGB2 NN BL GAM	β (SE)	β (SE)	edf
Vehicles per Person	N N N N N	-0.874*** (0.161)	[NA]	1.519***
Income per Capita	N N N Z N	-0.002 (0.002)	[NA]	3.851*
Number of Adults	P P P P P	0.014 (0.094)	0.025 (0.097)	[NA]
Number of Students	P P P P P	0.478* (0.146)	0.404** (0.151)	[NA]
Grocery Stores within 30 Minutes by Car	C Z N Z N	-0.004 (0.006)	[NA]	1.29
Grocery Stores within 30 Minutes by Walk	N Z N N N	-0.196 (0.185)	-0.187 (0.187)	[NA]
Grocery Stores Beyond Walk Access within 30 Minutes by Transit	P Z P P P	<u>0.421†</u> (0.234)	<u>0.409†</u> (0.237)	[NA]

*** <0.001, ** <0.01, * <0.05, † <0.1;

N = negative, P = positive, C = complex (notably non-monotonic), Z = negligible effect

Example Analysis | Trips Not Taken

Accumulated Local Effects (robust to collinearity)



SHAP
Values
with
LOESS Fit
Curve

XAI Data Viz Reference Guide

Your question	Reach for
Which features matter most?	Permutation importance · SHAP (mean value)
What shape is a feature's effect?	ALE (PDP if features are independent)
Do individuals respond differently?	ICE (centered)
Is there an interaction?	2-D PDP / ALE · fanning ICE lines
Why this one prediction?	SHAP waterfall · LIME
Smallest change to flip it?	Counterfactuals
Explain an image or text model?	Grad-CAM / saliency · attention

Three Takeaways

1

Assess Agreement

Agreement across varying models and data visualizations increases trust. Disagreement is information.

If disagreement is present, what is driving it?

2

Mind Correlation

Correlated features can mislead. Understand which data visualization techniques are robust to collinearity.

Consider feature engineering via feature selection and/or dimensionality reduction.

3

Explanation $\neq$ Causation

Explainable data visualizations explain the model, not the world. Assessing causality requires experimental design.

Has other literature already established causal mechanisms?

Agentic Documentation Pipeline

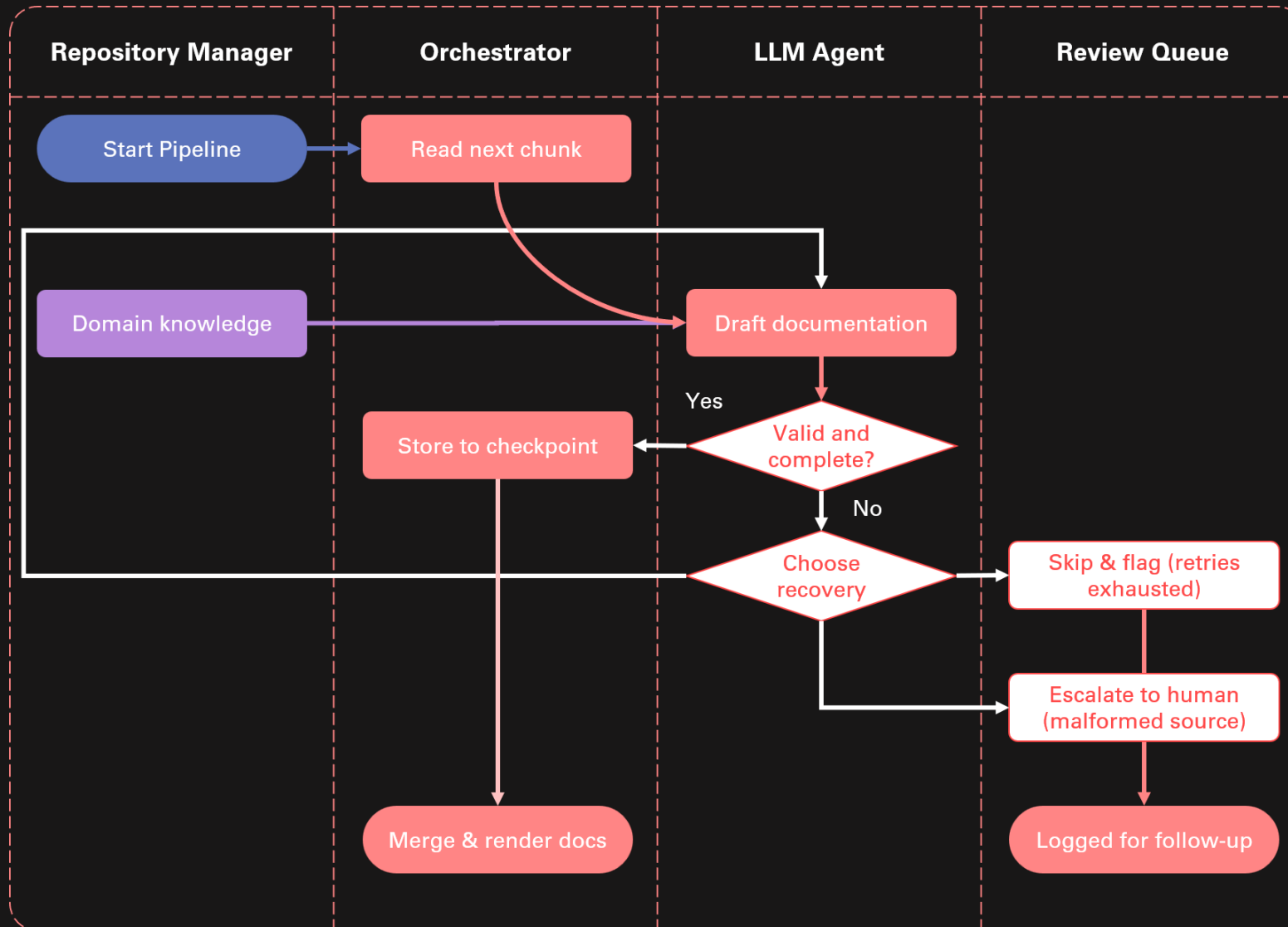
Why an Agentic Pipeline?

- Travel Demand Models are large, complex repositories.
- Creating and maintaining model documentation takes a large amount of expert person hours.
- LLMs (even with longer context windows), by themselves, struggle with attention and hallucination.

Advantages of an Agentic Workflow

- LLM only sees a part of the codebase at a time and documents the chunk.
- LLM interprets the error in the pipeline and decides how to act on it.

TRMG2 Documentation Agent



Color Legend -----

Agentic

Programmatic

Human

Multisource

Shape Legend -----

Start/End Point

Process

Decision Point

Web Documentation Interface

TRMG2 documentation

GitHub

@ 328678a

Part 1: Architecture Overview and Model Structure

1. What TRMG2 Is2 files

2. Directory Structure2 files

3. Model Execution Flow2 files

4. Key Design Principles1 file

5. Time Periods and Mode Codes1 file

6. Core Data Files

7. GISDK Language Conventions

8. Args Object

Part 2: Socioeconomic Data Pipeline, Population Synthesis, and Trip Generation

1. Socioeconomic Data Preparation2 files

1.1 SE Master File2 files

1.2 SE Validation (Step 02)1 file

2. Network Preprocessing (Step 02)1 file

3. Accessibility Computation (Step 03)1 file

4. Population Synthesis (Step 04)2 files

5. Trip Generation (Step 05)3 files

6. Nonmotorized Model (Step 06)3 files

7. Aggregation and TOD (Steps 07–08)2 files

Part 3: Skimming, Parking, Mode Choice, and Destination Choice

Part 4: Assignment, Special Generators, and External Models

Part 1: Architecture Overview and Model Structure > 1. What TRMG2 Is

1. What TRMG2 Is

Airport.rscCommercial Vehicles.rsc

TRMG2 is a trip-based (4-step) travel demand model implemented entirely in TransCAD's GISDK scripting language. It replaces TRMG1 (the prior TransCAD-based TRM) and serves as the official planning model for the Research Triangle region's two MPOs. The codebase is open source and version-controlled on GitHub.

The model covers the full urban modeling workflow: socioeconomic data disaggregation and population synthesis, trip generation, nonmotorized mode split, time-of-day distribution, destination choice, mode choice, network assignment, and comprehensive validation summaries. Special generators for airport, commercial vehicles, university travel, and external trips supplement the resident model.

src/Airport.rsc

339 lines · 10,492 chars

Find in file...

View on GitHub

1 /*

2 Called by flowchart

3

4 Generates and distributes airport trips. Because the models are simple and

5 do not depend on congested times, it can be done outside the feedback loop.

6 Mode choice for airport trips happens within the feedback loop and uses the

7 N_HB_OD_Long_vs probabilities.

8 */

9

10 Macro "Airport" (Args)

11 RunMacro("Airport Productions", Args)

12 RunMacro("Airport Distribution", Args)

13 RunMacro("Airport TOD", Args)

14 return(1)

15 endmacro

16

17 /*

18 Called by flowchart

19 */

20

21 Macro "Airport MC" (Args)

22 RunMacro("Airport Mode Choice", Args)

23 RunMacro("Airport Separate Auto and Transit", Args)

24 RunMacro("Airport Directionality", Args)

25 return(1)

26 endmacro

27

28 /*

29 Calculate airport productions

30 */

31

32 Macro "Airport Productions" (Args)

33 se_file = Args.SE

34 skim_dir = Args.[Output Folder] + "\\skims\\roadway"

35 airport_model_file = Args.[Input Folder] + "\\airport\\airport_model.csv"

36

37 se_vw = OpenTable("se", "FFB", {se_file})

38

39 data = GetDataVectors(

Files: 55Sections: 241Viewing: src/Airport.rsc

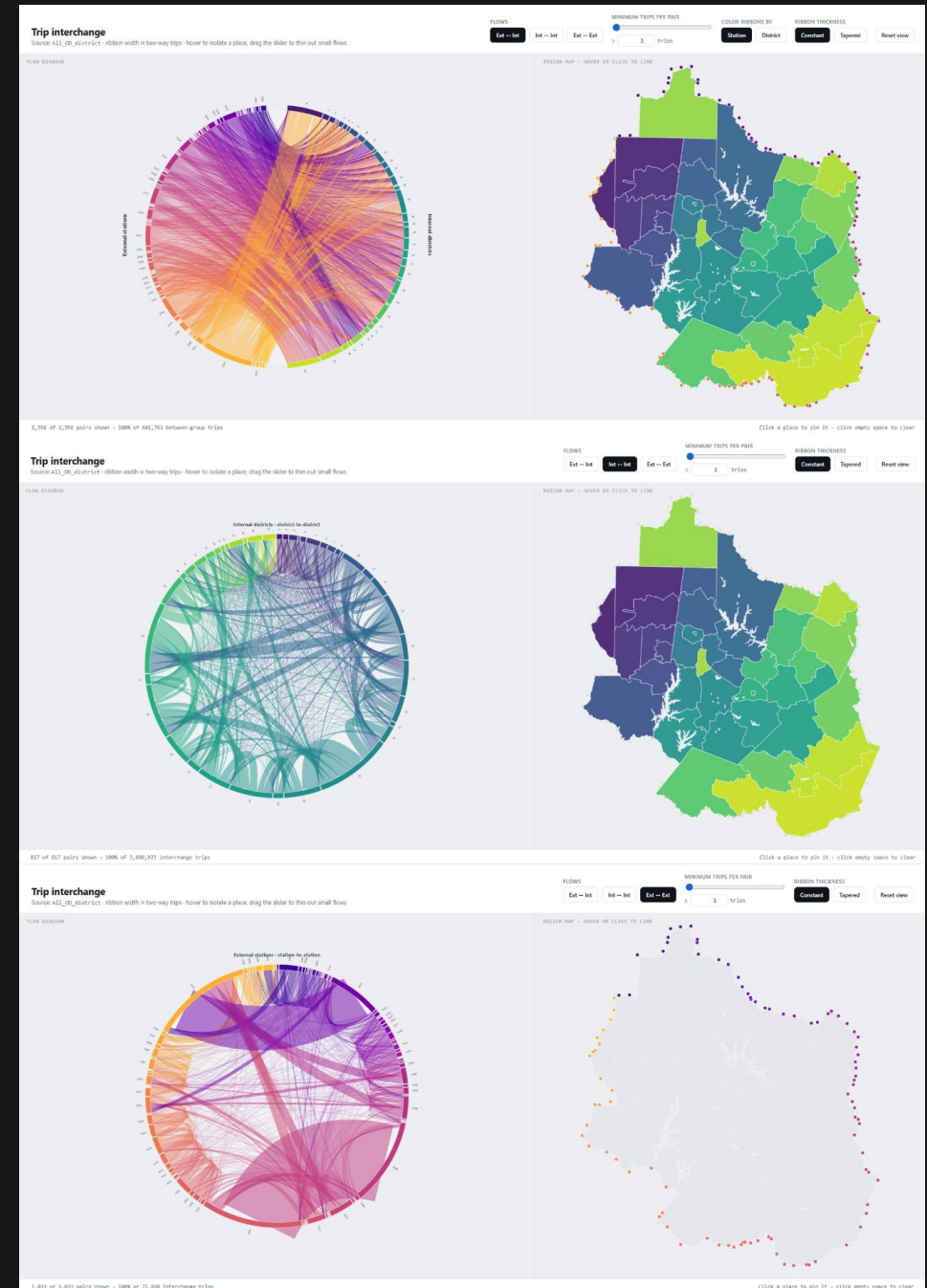
#n=part1-h2-1-what-trmg2-is&f=src%2FAirport.rsc

Agentic Documentation Next Steps

- Validation with larger group of model users
- Make the pipeline end-to-end (currently split: docx generation and website rendering)
- Experiment with:
 - developing a RAG chatbot on top of the documentation
 - creating knowledge graph and ontology for quality assurance and reasoning with documentation
 - providing more agency to the agent(s): planning, organizing, and decisions
 - developing custom harness for improved quality
 - and more

Other AI Use Cases

- Interactive data dashboards.
- Sophisticated data visualization with code auditability.
- Imputation, estimation, and prediction.
 - Imputing K-12 enrollment.
 - Imputing missing trip/visit observations from LBS datasets.
 - Estimating and predicting bicycle and pedestrian trips.
- Enhanced systematic literature review workflows (LLM joint screening)
- NLP processing and semantic extraction.
- VLM-based extraction for transforming unstructured land use documents into structured data tables.



Institute for Transportation Research and Education at NC State University



- Aviation
- Highway Systems
- Bicycle and Pedestrian
- Port and Ferry
- Rail
- Transit
- School Planning and Transportation
- Systems Planning and Analysis

Contact Information



Alexander Yoshizumi
Program Manager – Systems Planning
& Analysis
ayoshiz@ncsu.edu



Si Shi
Research Scholar and TRM Lead
Modeler
sshi7@ncsu.edu



Kshitiz Khanal
Research Scholar and TRM Modeler
kkhanal2@ncsu.edu



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